

The Transformer Architecture & Self-Attention

The 2017 Breakthrough: 'Attention Is All You Need'

The Transformer architecture, introduced in the seminal 2017 Google Brain paper, replaced Recurrent Neural Networks (RNNs) as the dominant NLP architecture. The key innovation: self-attention, which allows every token to directly interact with every other token in a sequence.

RNN limitation: sequential processing means early context is 'forgotten' in long sequences. Transformers solve this with self-attention.

Self-Attention: Query, Key, Value

Query (Q): The token asking 'what should I pay attention to?'

Key (K): Each other token broadcasts 'here's what I represent'.

Value (V): The actual information to be aggregated once attention weights are computed.

For each token, self-attention: (1) computes similarity between its Q and all Ks, (2) applies softmax to get attention weights, (3) takes a weighted sum of Vs.

Multi-Head Attention

Running self-attention once would only capture one type of relationship. Multi-head attention runs several attention operations in parallel, each potentially focusing on a different aspect (syntax, semantics, coreference). The outputs are concatenated and projected.

Example: in the sentence 'The animal didn't cross the street because it was too tired', self-attention links 'it' to 'animal' — resolving coreference.

Positional Encoding

Unlike RNNs, Transformers process all tokens in parallel — they have no built-in sense of word order. Positional encodings (sinusoidal functions or learned vectors) are added to embeddings so the model knows token position.

The Encoder-Decoder Structure

Encoder-only (BERT): Reads the full sequence — used for classification, question answering, embeddings.

Decoder-only (GPT, Claude): Generates text auto-regressively — predicts the next token, then the next, etc.

Encoder-Decoder (T5, BART): Encodes a source sequence, then decodes a target — used for translation and summarisation.

Why Transformers Dominate

- Parallelisation: all tokens processed simultaneously — massively faster training.
- Long-range dependencies: self-attention connects any two tokens in one step regardless of distance.
- Scalability: performance scales predictably with data, parameters, and compute (scaling laws).
- Transfer learning: pre-train once, fine-tune for hundreds of downstream tasks.

Key Takeaways

- The Transformer (2017) replaced RNNs by using self-attention instead of recurrence.
- Self-attention uses Q/K/V to let every token attend to every other token.
- Multi-head attention captures multiple relationship types in parallel.
- Three variants: Encoder-only, Decoder-only, Encoder-Decoder.

Quick-Revision Questions

- Explain the Q, K, V mechanism in self-attention using an analogy.
- Why do Transformers need positional encodings?
- What variant of Transformer does GPT use, and why?